## FCFS

```
#include <stdio.h>
#include <stdlib.h>
int main() {
int n, head, i;
int queue[20];
int totalSeek = 0;
printf("FCFS Disk Scheduling\n");
printf("Enter number of requests: ");
scanf("%d", &n);
printf("Enter request queue:\n");
for(i = 0; i < n; i++) {
scanf("%d", &queue[i]);
}
printf("Enter initial head position: ");
scanf("%d", &head);
printf("\nSeek Sequence:\n");
printf("%d ", head);
for(i = 0; i < n; i++) {
totalSeek += abs(queue[i] - head);
head = queue[i];
printf("-> %d ", head);
}
printf("\n\nTotal Seek Time = %d\n", totalSeek);
return 0;
}
```

```
pratyushojha@Pratyushs-MacBook-Pro Coding % cd "/Users/pratyushojha/Documents/Coding/" && gcc 10a.c -o 10a && "
/Users/pratyushojha/Documents/Coding/"10a
1WA24CS216
FCFS Disk Scheduling
Enter number of requests: 8
Enter request queue:
98 183 37 122 14 124 65 67
Enter initial head position: 53

Seek Sequence:
53 -> 98 -> 183 -> 37 -> 122 -> 14 -> 124 -> 65 -> 67

Total Seek Time = 640
pratyushojha@Pratyushs-MacBook-Pro Coding %
```

## SCAN

```
#include <stdio.h>
#include <stdlib.h>
void sort(int arr[], int n) {
int i, j, temp;
for(i = 0; i < n - 1; i++) {
for(j = 0; j < n - i - 1; j++) {
if(arr[j] > arr[j + 1]) {
temp = arr[j];
arr[j] = arr[j + 1];
arr[j + 1] = temp;
}
}
}
}
int main() {
int queue[20], n, head, diskSize;
int i, j;
int totalSeek = 0;
int tempHead;
printf("SCAN Disk Scheduling\n");
printf("Enter number of requests: ");scanf("%d", &n);
printf("Enter request queue:\n");
for(i = 0; i < n; i++) {
scanf("%d", &queue[i]);
}
printf("Enter initial head position: ");
scanf("%d", &head
printf("Enter disk size: ");
scanf("%d", &diskSize);
if(head < 0 || head >= diskSize) {
printf("Invalid head position\n");
return 0;
}
for(i = 0; i < n; i++) {
if(queue[i] < 0 || queue[i] >= diskSize) {
printf("Invalid disk request: %d\n", queue[i]);
return 0;
}
}
sort(queue, n);
tempHead = head;
printf("\nSeek Sequence:\n");
printf("%d", head);
for(i = 0; i < n; i++) {
if(queue[i] >= head)
break;
}
for(j = i; j < n; j++) {
totalSeek += abs(queue[j] - tempHead);
tempHead = queue[j];
printf(" -> %d", tempHead);
}
if(tempHead != diskSize - 1) {
totalSeek += abs((diskSize - 1) - tempHead);
```

```
tempHead = diskSize - 1;
printf(" -> %d", tempHead);
}
for(j = i - 1; j >= 0; j--) {
totalSeek  += abs(queue[j] - tempHead);
tempHead = queue[j];printf("  -> %d", tempHead);
}
printf("\n\nTotal  Seek Time = %d\n", totalSeek);
return  0;
}
```

```
pratyushojha@Pratyushs-MacBook-Pro Coding % cd "/Users/pratyushojha/Documents/Coding/" && gcc 10b.c -o 10b && "
/Users/pratyushojha/Documents/Coding/"10b
1WA24CS216
SCAN Disk Scheduling
Enter number of requests: 8
Enter request queue:
98 183 37 122 14 124 65 67
Enter initial head position: 53
Enter disk size: 200

Seek Sequence:
53 -> 65 -> 67 -> 98 -> 122 -> 124 -> 183 -> 199 -> 37 -> 14

Total Seek Time = 331
pratyushojha@Pratyushs-MacBook-Pro Coding %
```

## SSTF

```
#include <stdio.h>
#include <stdlib.h>
int main() {
int queue[20], visited[20] = {0};
```

```c
int n, head, i, j;
int totalSeek = 0;
int tempHead;
printf("SSTF Disk Scheduling\n");
printf("Enter number of requests: ");
scanf("%d", &n);
printf("Enter request queue:\n");
for(i = 0; i < n; i++) {
scanf("%d", &queue[i]);
}
printf("Enter initial head position: ");scanf("%d", &head);
tempHead = head;
printf("\nSeek Sequence:\n");
printf("%d", tempHead);
for(i = 0; i < n; i++) {
int min = 9999;
int index = -1;
for(j = 0; j < n; j++) {
if(visited[j] == 0) {
int distance = abs(queue[j] - tempHead);
if(distance < min) {
min = distance;
index = j;
}
}
}
visited[index] = 1;
totalSeek += min;
tempHead = queue[index];
printf(" -> %d", tempHead);
}
printf("\n\nTotal Seek Time = %d\n", totalSeek);
return 0;
}
```

```
pratyushojha@Pratyushs-MacBook-Pro Coding % cd "/Users/pratyushojha/Documents/Coding/" && gcc 10c.c -o 10c && "
/Users/pratyushojha/Documents/Coding/"10c
1WA24CS216
SSTF Disk Scheduling
Enter number of requests: 8
Enter request queue:
98 183 37 122 14 124 65 67
Enter initial head position: 53

Seek Sequence:
53 -> 65 -> 67 -> 37 -> 14 -> 98 -> 122 -> 124 -> 183

Total Seek Time = 236
pratyushojha@Pratyushs-MacBook-Pro Coding %
```

## C-LOOK

```
#include <stdio.h>
#include <stdlib.h>
void sort(int arr[], int n) {
int i, j, temp;
for(i = 0; i < n - 1; i++) {
for(j = 0; j < n - i - 1; j++) {
if(arr[j] > arr[j + 1]) {
temp = arr[j];
arr[j] = arr[j + 1];
arr[j + 1] = temp;
}
}
}
}
int main() {
int queue[20], n, head;
int i, j;
int totalSeek = 0;
int tempHead;
```

```
int index = 0;
printf("C-LOOK Disk Scheduling\n");
printf("Enter number of requests: ");
scanf("%d", &n);
printf("Enter request queue:\n");
for(i = 0; i < n; i++) {
scanf("%d", &queue[i]);
}
printf("Enter initial head position: ");
scanf("%d", &head);
sort(queue, n);
tempHead = head;
printf("\nSeek Sequence:\n");
printf("%d", tempHead);
for(i = 0; i < n; i++) {
if(queue[i] >= tempHead) {
index = i;
break;
}}
for(j = index; j < n; j++) {
totalSeek += abs(queue[j] - tempHead);
tempHead = queue[j];
printf(" -> %d", tempHead);
}
if(index != 0) {
totalSeek += abs(tempHead - queue[0]);
tempHead = queue[0];
printf(" -> %d", tempHead);
}
for(j = 1; j < index; j++) {
totalSeek += abs(queue[j] - tempHead);
tempHead = queue[j];
printf(" -> %d", tempHead);
}
printf("\n\nTotal Seek Time = %d\n", totalSeek);
return 0;
}
```

```
pratyushojha@Pratyushs-MacBook-Pro Coding % cd "/Users/pratyushojha/Documents/Coding/" && gcc 10d.c -o 10d && "
/Users/pratyushojha/Documents/Coding/"10d
1WA24CS216
C-LOOK Disk Scheduling
Enter number of requests: 8
Enter request queue:
98 183 37 122 14 124 65 67
Enter initial head position: 53

Seek Sequence:
53 -> 65 -> 67 -> 98 -> 122 -> 124 -> 183 -> 14 -> 37

Total Seek Time = 322
pratyushojha@Pratyushs-MacBook-Pro Coding %
```